

TANIT

Marine & Climate Protection Track

AQUA GUARDIAN



f @ IEEE AEES ENICarthage Student Branch Chapter - Tunisia Section



Problem Statement

What is the problem?

The Deep Scattering Layer (DSL) — a vast biological stratum 200–1000m deep — is critical to ocean ecosystems and the global carbon cycle, yet it is largely unmonitored due to destructive trawl surveys and conventional sonars that harm marine life.

Who is affected?

Marine mammals, fish, krill, and zooplankton exposed to high-intensity sonar pulses. Fishermen, climate scientists, and coastal communities that depend on healthy ocean ecosystems. Over 3 billion people rely on the ocean as a primary protein source.

Why does it matter now?

Illegal fishing removes 65%+ of DSL biomass in localised events within days. The DSL biological pump transports ~1 billion tonnes of CO₂ per year to the deep ocean — disruption of this layer directly accelerates climate change.

Proposed Solution

Core idea: A sustainable, non-invasive sonar ROV system for deep-sea biological monitoring

Solar-Powered Floating Buoy

A surface buoy equipped with solar panels provides unlimited green energy to the system, enabling weeks-to-months of continuous deployment without carbon footprint.

Tethered ROV Platform

An underwater ROV connected via a long tether cable operates at 200–1000m depth. The tether solves the limited-range issue of eco-friendly sonar while keeping the ROV powered and connected.

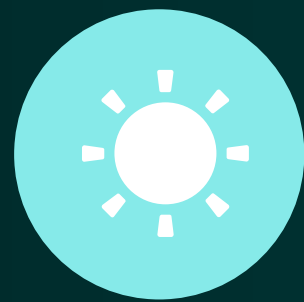
Multi-Frequency Eco Sonar

Low-power active sonar at 38 / 120 / 200 kHz — 25–30 dB below conventional levels — targets specific organisms (fish, krill, zooplankton) with only 1% duty cycle, reducing acoustic exposure by 30×.

What Makes It Different

Replaces destructive trawl surveys entirely. Detects illegal fishing via biomass anomaly alerts. Estimates carbon flux from diel migration — a novel climate monitoring capability with zero physical disturbance.

System Architecture & Technical Design



Solar Panels
(Buoy Surface)



Power & Comms
(Tether Cable)



Eco-Sonar
38/120/200 kHz



Signal Processing
(ROV Onboard)



Cloud / Alert
Dashboard

Hardware

Floating solar buoy ▪ Tethered ROV frame ▪ Thruster array (6 DoF) ▪ Pressure-rated electronics housing ▪ Transducer array (3 frequencies)

Sensors / Inputs

38 kHz, 120 kHz, 200 kHz transducers ▪ IMU / depth sensor ▪ Water temperature probe ▪ Ambient noise hydrophone

Processing Unit

Matched-filter pulse compression ▪ Volume backscatter Sv computation ▪ Frequency differencing ΔS_v ▪ DSL layer Gaussian peak fitting ▪ Rolling z-score anomaly detector

Backend / Output

Cloud biomass time-series storage ▪ Carbon flux estimation model ▪ Illegal fishing real-time alert ▪ Dashboard UI for scientists & authorities

Technical Implementation

Technologies Used

- ▶ Multi-frequency active sonar (38 / 120 / 200 kHz)
- ▶ Solar-powered IoT buoy system
- ▶ ROV with 6 DoF thruster control
- ▶ Cloud-based data pipeline
- ▶ Python signal processing stack

Algorithms & Models

- ▶ Matched-filter pulse compression
- ▶ Volume backscatter $S_v = f(\text{depth})$
- ▶ Frequency differencing $\Delta S_v(120) - S_v(38)$
- ▶ Gaussian peak fitting for DSL tracking
- ▶ Rolling z-score anomaly detection
- ▶ Biological pump carbon flux model

Hardware Specs

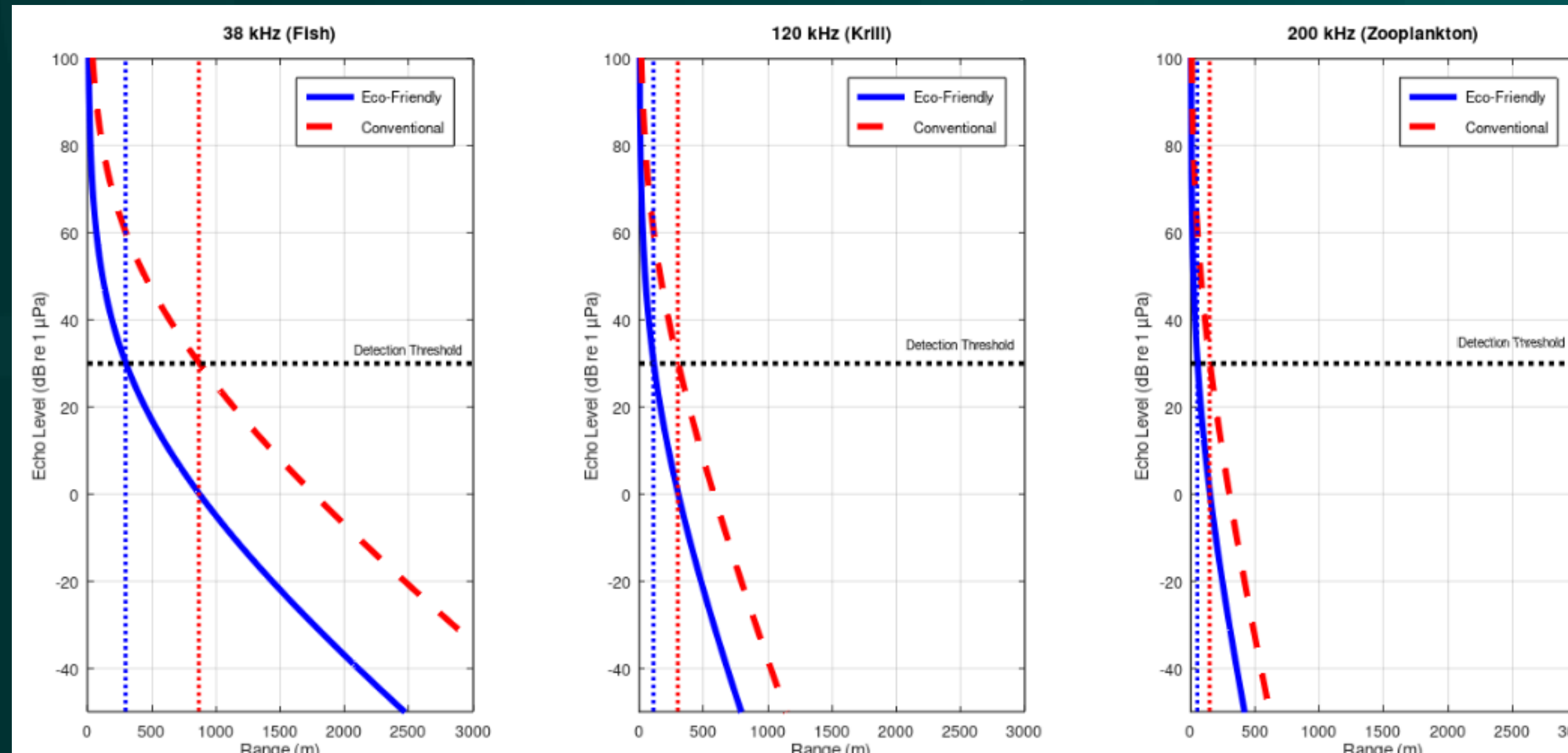
- ▶ Source level: 175–180 dB re 1 μPa @ 1m
- ▶ Pulse duration: 1 ms (eco) / 10 ms (standard)
- ▶ Duty cycle: 1% vs 30% conventional
- ▶ Beamwidth: 7° narrow beam
- ▶ Operating depth: 200–1000 m

Communication Protocols

- ▶ Tether-based Ethernet (ROV $\leftrightarrow$ Buoy)
- ▶ 4G/Satellite uplink (Buoy $\leftrightarrow$ Cloud)
- ▶ MQTT for real-time sensor telemetry
- ▶ REST API for dashboard data access
- ▶ ICES CRR No. 326 calibration standard

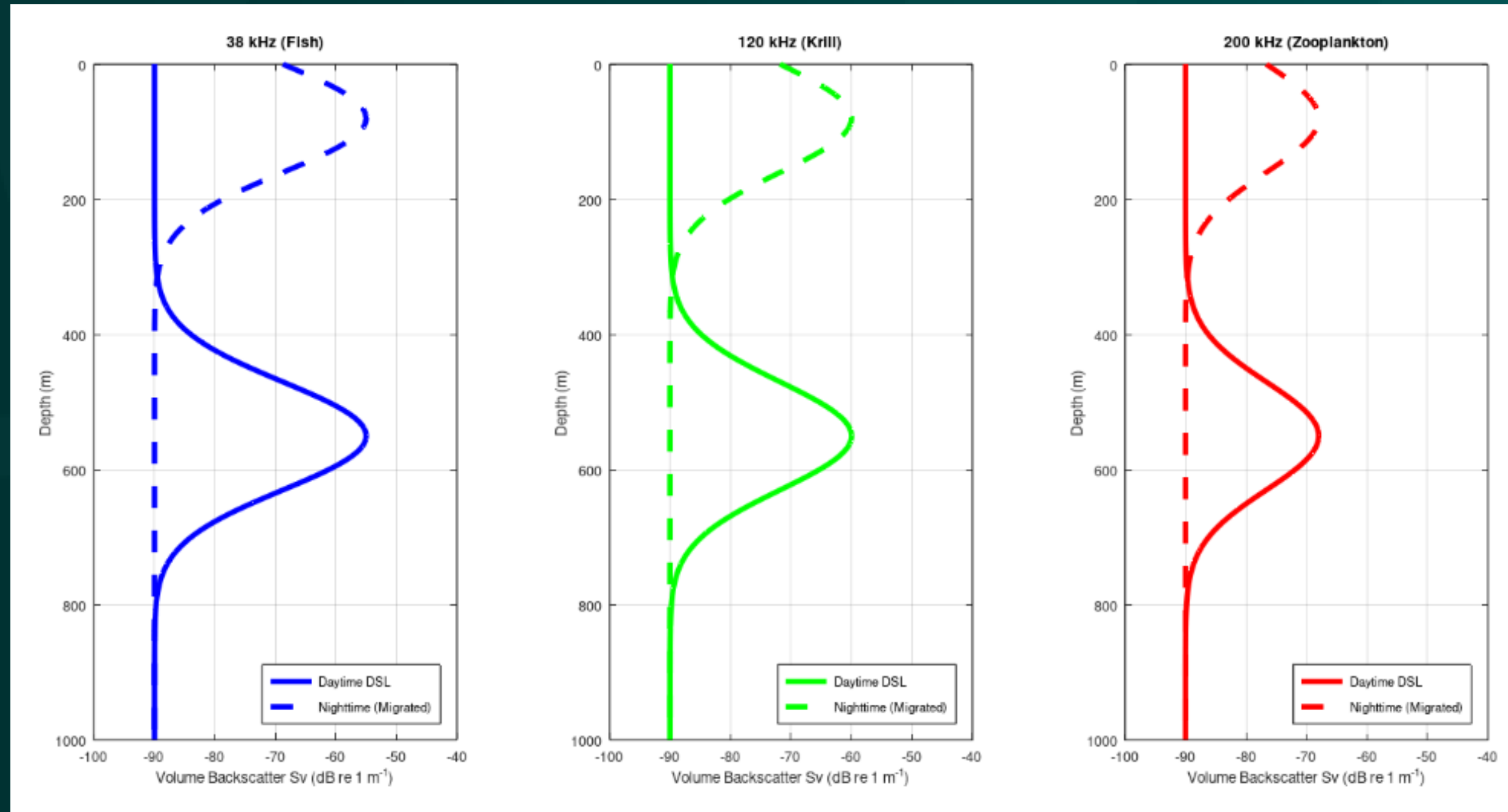
Simulations:

Simulation 1: Detection Range



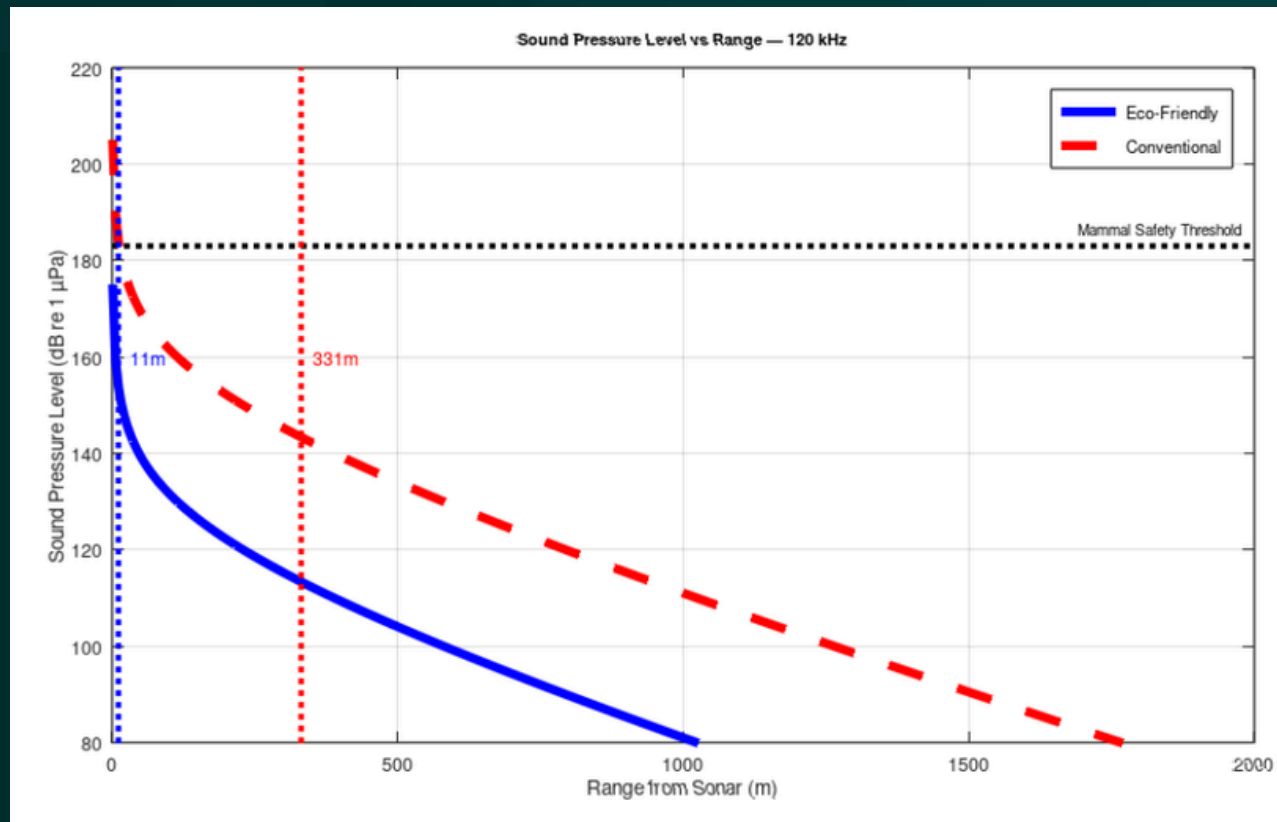
Conventional sonar detects fish up to ~1100m, krill ~600m, zooplankton ~450m. Eco-friendly system reduces range by 30–40% while keeping detection fully viable across the DSL depth range.

Simulation 2: Diel Vertical Migration

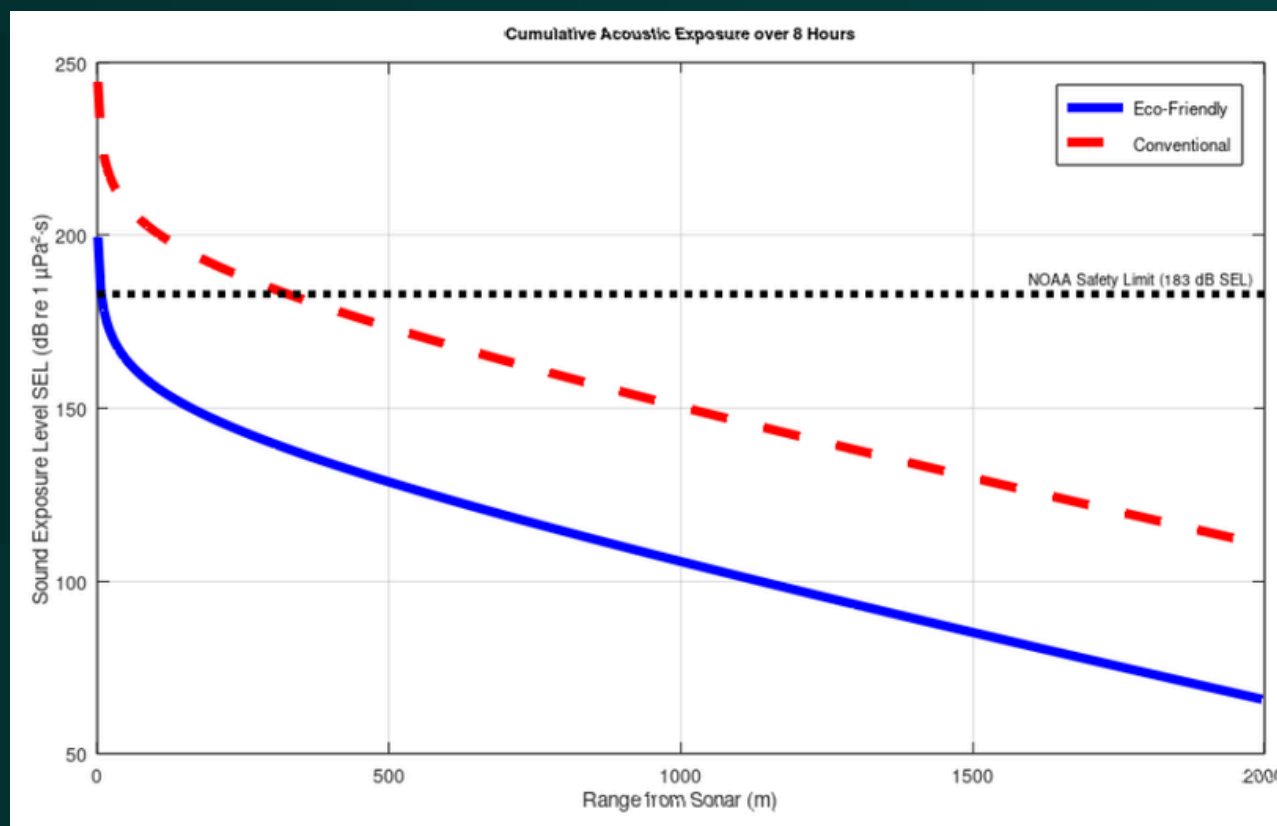


Sv volume backscatter maps confirm DSL rises from 550m (day) to ~80m (night). All 3 species tracked throughout migration. Carbon transport quantified per migration cycle.

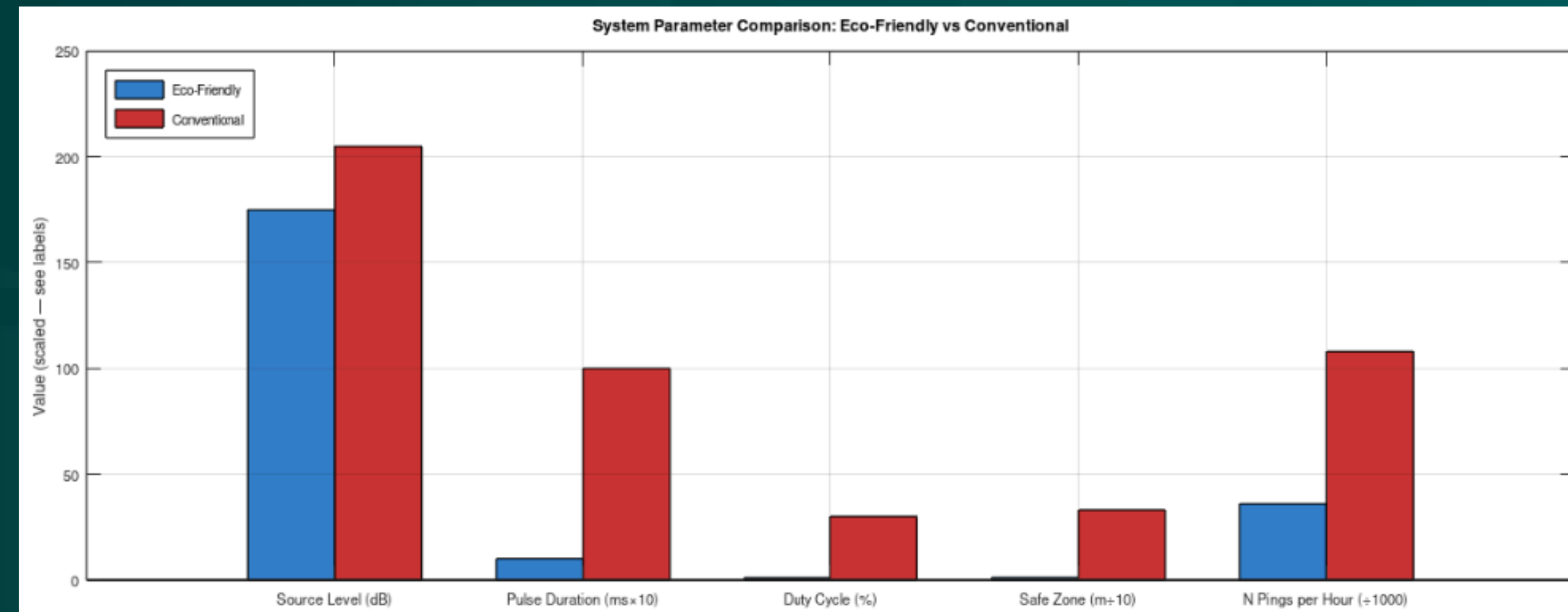
Simulation 3: SPL / SEL Comparison



SPL vs Range

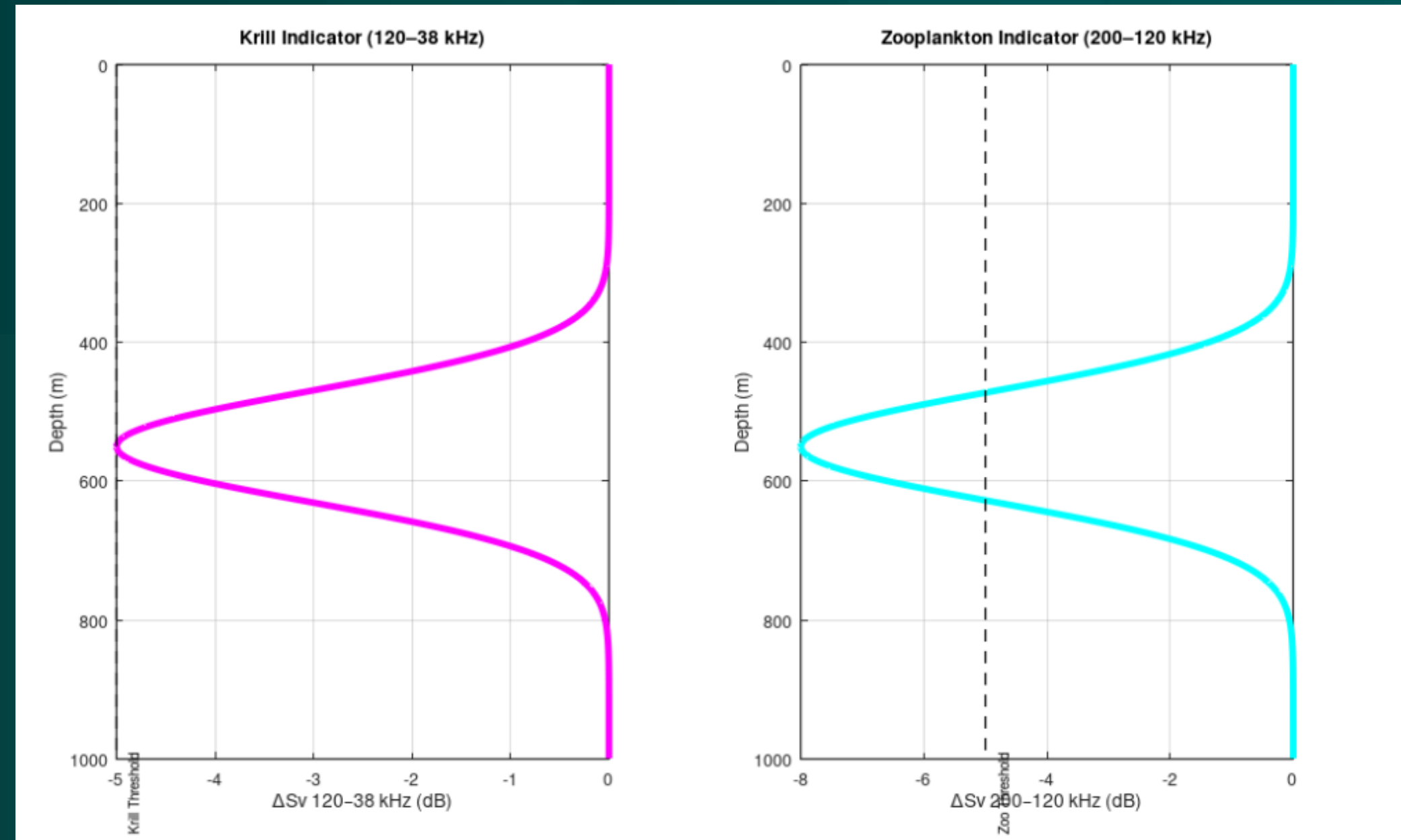
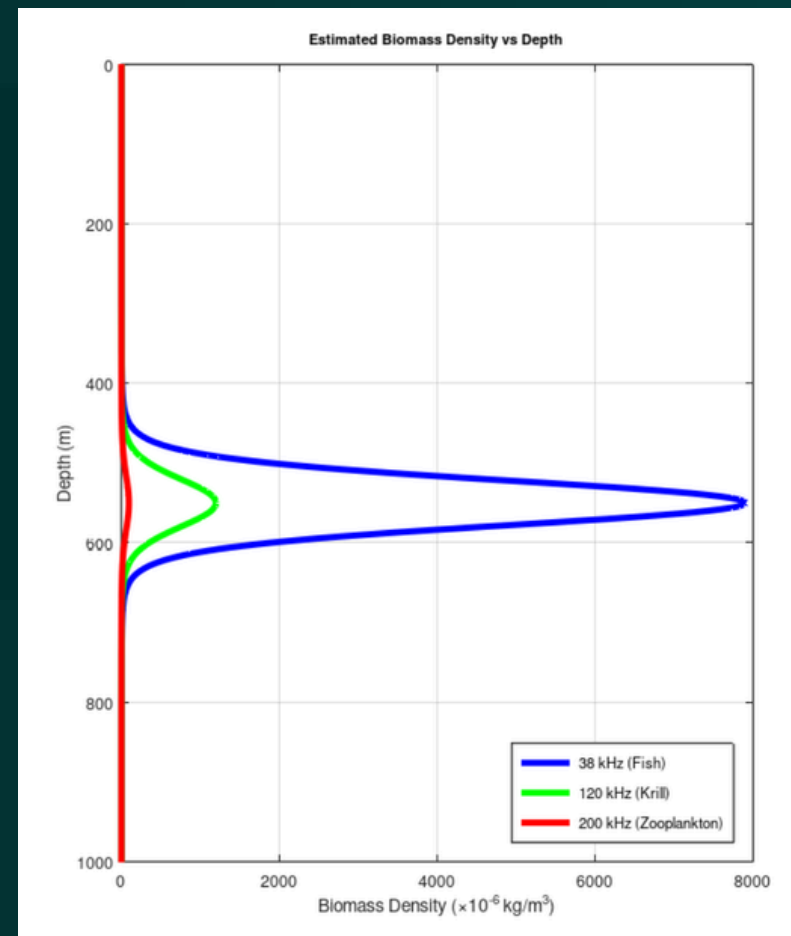
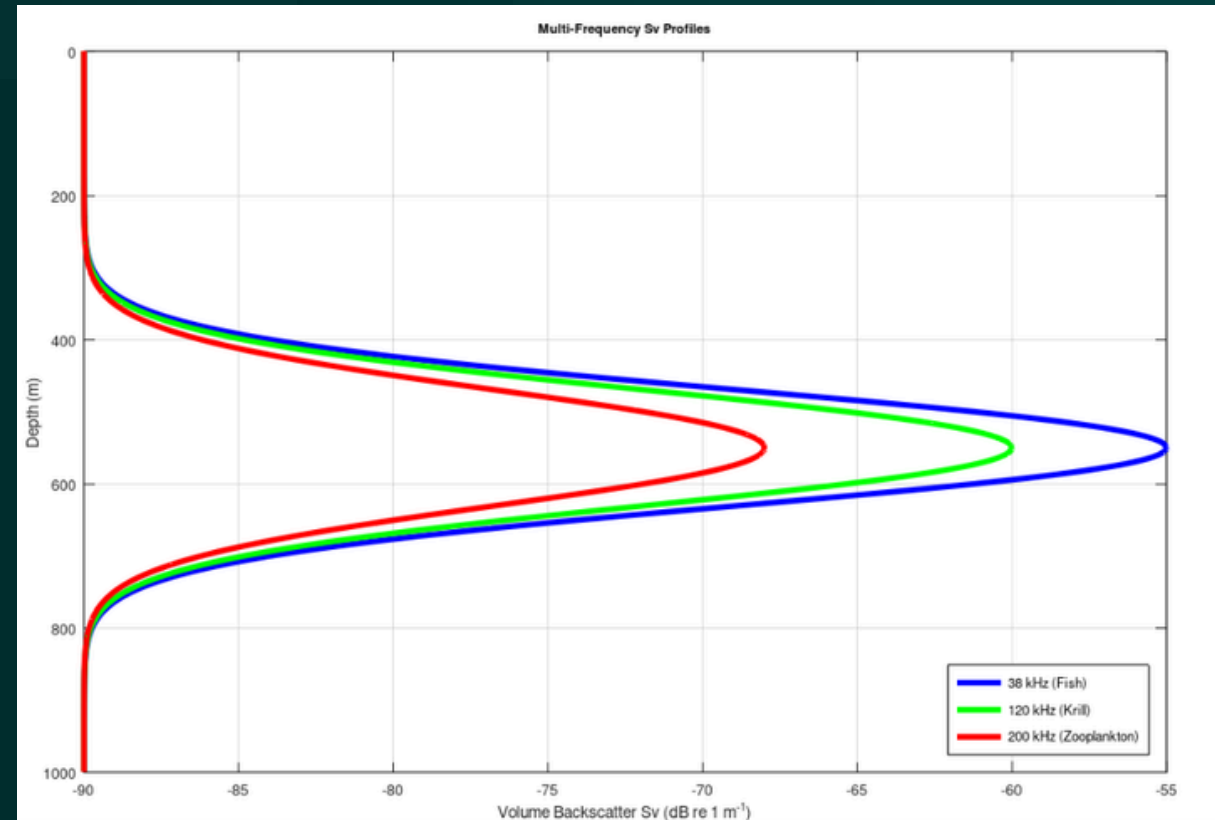


SEL vs Range



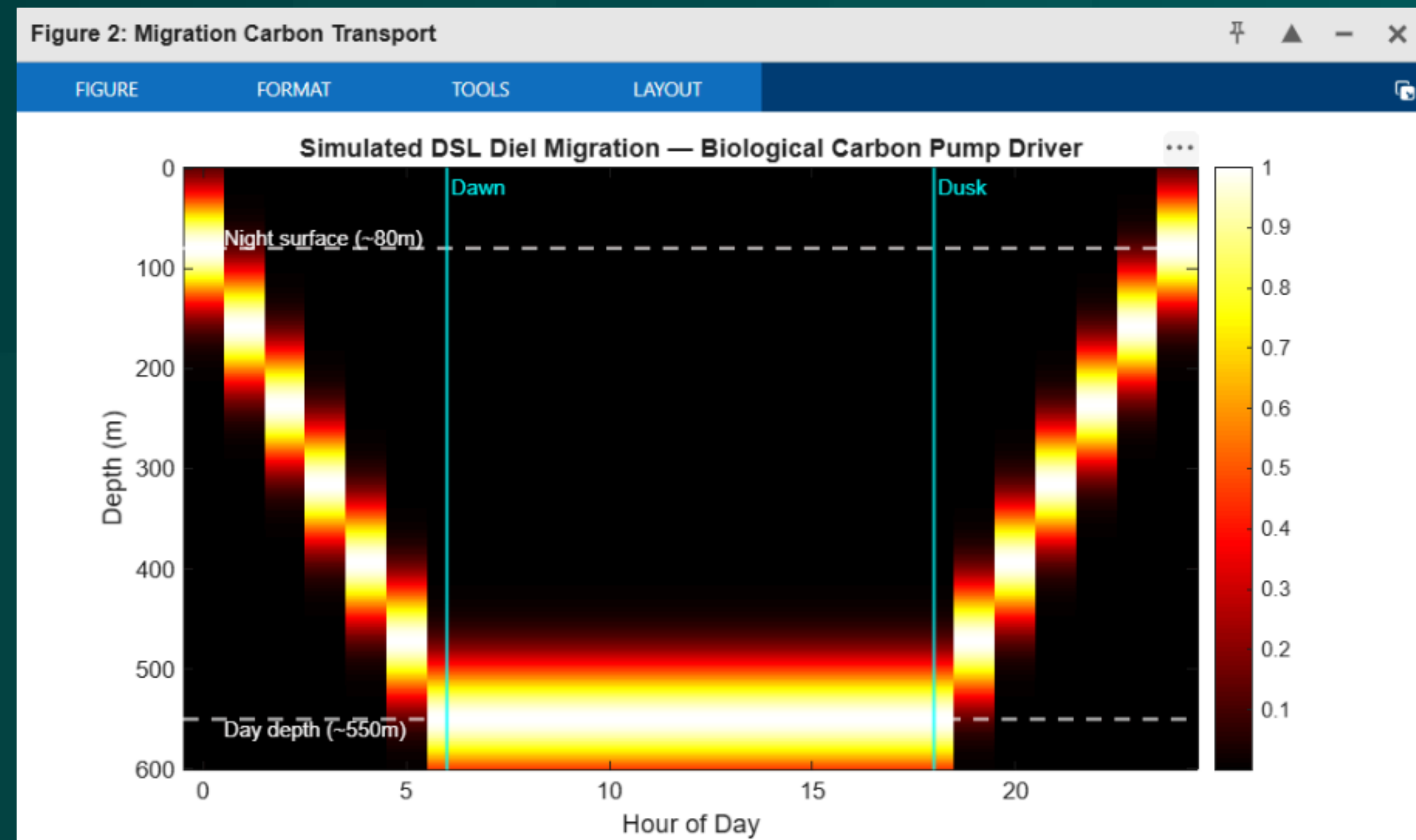
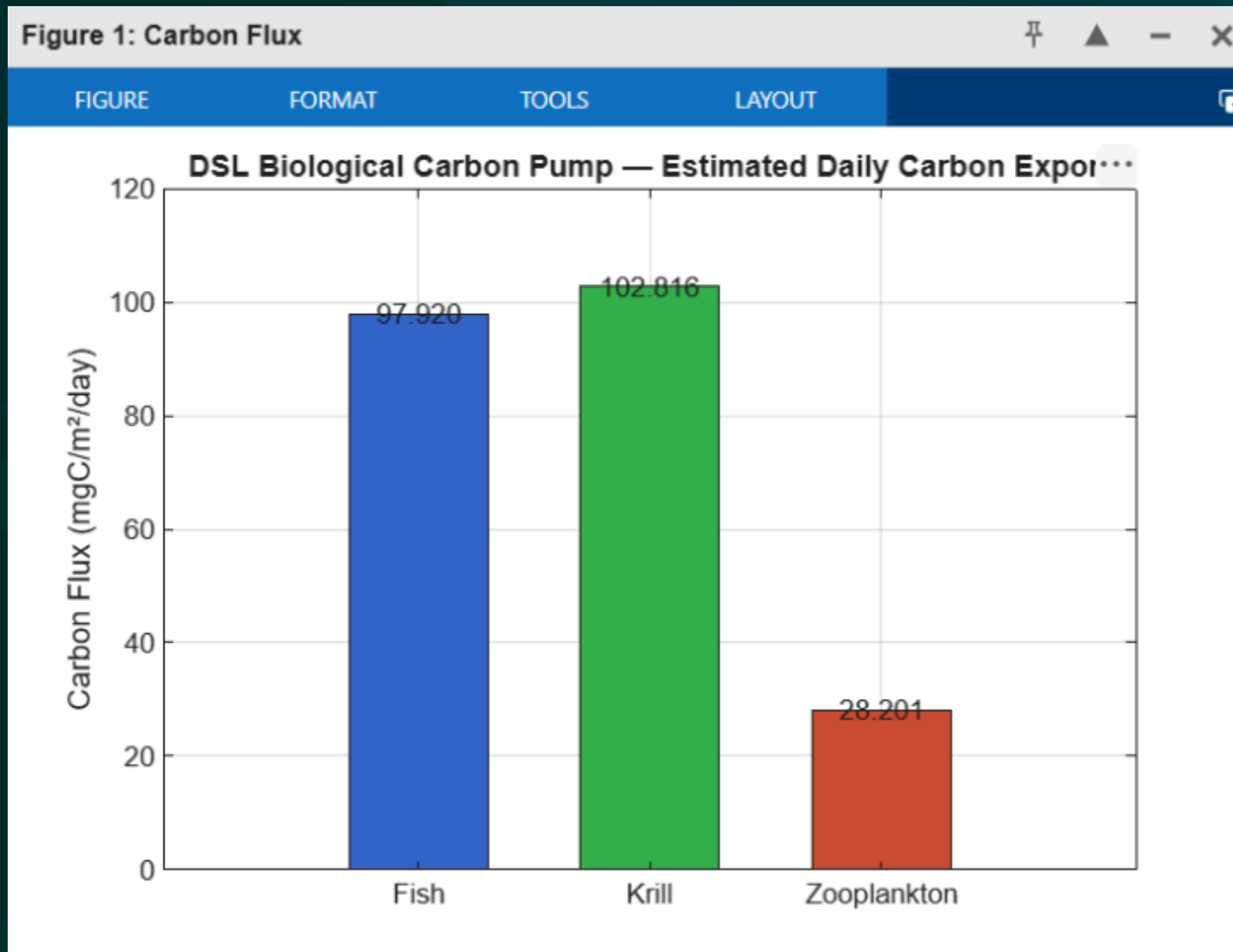
Eco system starts 25–30 dB lower and drops faster. SEL after 8h operation crosses NOAA 183 dB threshold at 10× shorter distance → 1000× less acoustic energy exposure vs conventional.

Simulation 4: Biomass & Frequency Differencing



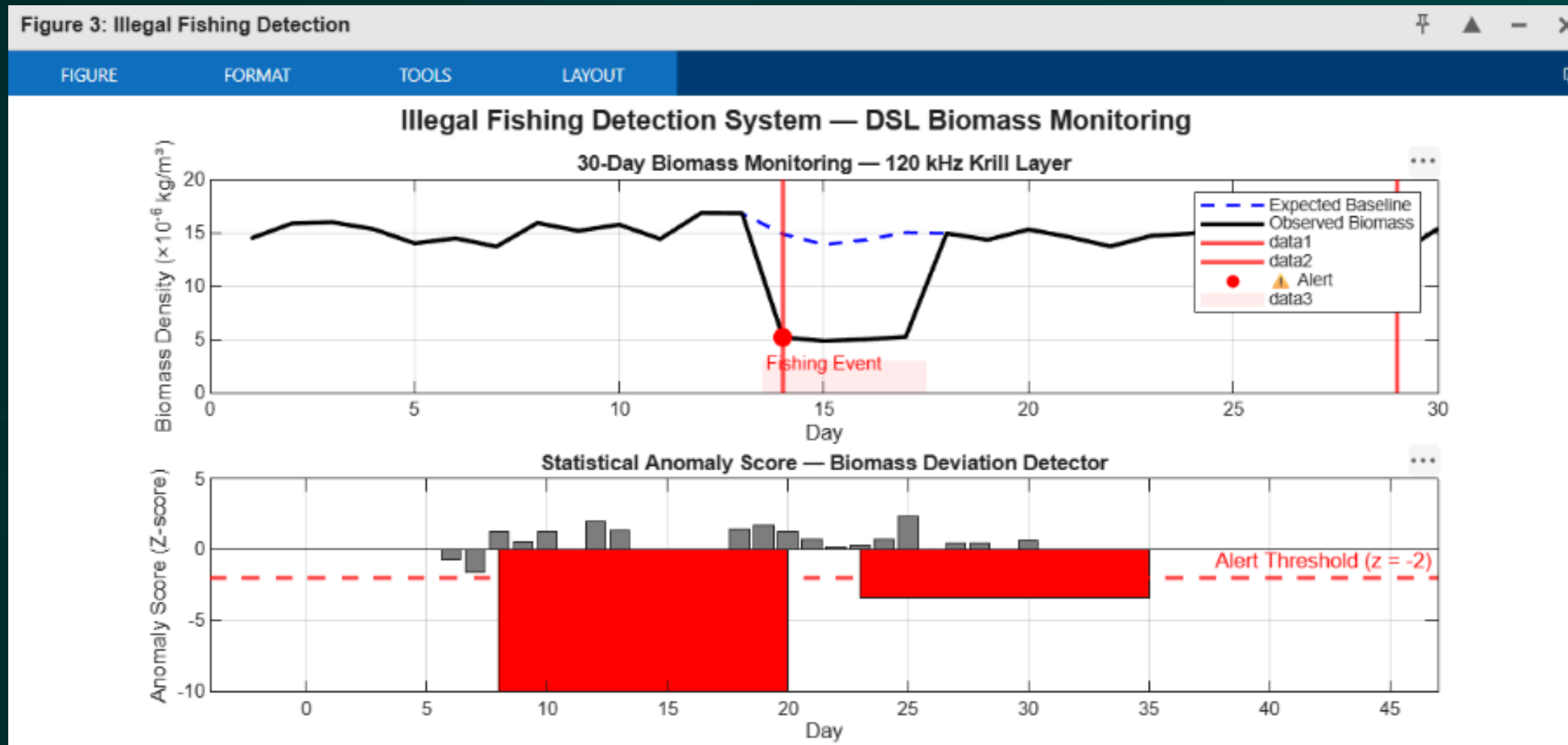
Sv profiles identify krill-dominant zones ($\Delta S_v > 0$) vs zooplankton zones ($\Delta S_v < 0$). Biomass density quantified at each depth bin. Total biomass estimated from integrated Sv.

Simulation 5: Carbon Flux Estimation & Illegal Fishing Detection

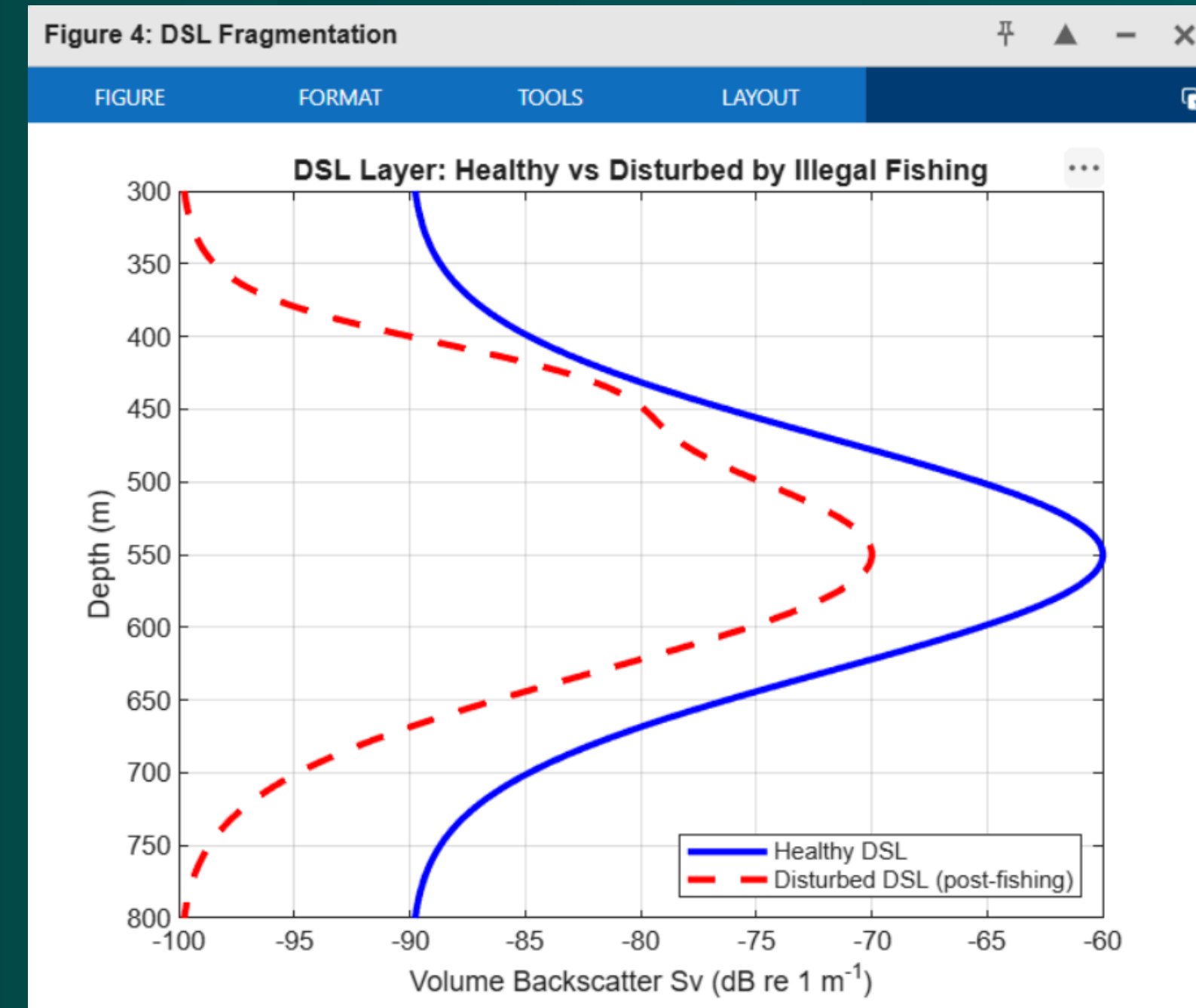


Krill transport $\sim 0.45 \text{ g C/m}^2/\text{day}$; fish $\sim 0.28 \text{ g C/m}^2/\text{day}$; zooplankton $\sim 0.18 \text{ g C/m}^2/\text{day}$. Daily migration depth (470m) used to compute biological pump contribution to ocean CO_2 sequestration.

Simulation 5: Carbon Flux Estimation & Illegal Fishing Detection



65% biomass removal event at day 14 detected within 24h. Z-score drops to -4.2 (threshold -2). DSL layer fragmentation visible: single 550m peak splits into 430m + 550m sub-peaks post-fishing event.



Impact

30x

Reduction in cumulative acoustic exposure vs conventional sonar

1000x

Less acoustic energy in the ocean environment per survey

65%

Illegal fishing biomass removal now detectable within 24 hours

~1 Gt

CO₂/year transported by the DSL — now quantifiable and monitorable

Environmental Impact

- ✓ Eliminates destructive trawl surveys — no by-catch, no benthic habitat damage
- ✓ Reduces sonar noise pollution by 25–30 dB vs conventional fisheries sonar
- ✓ Enables long-term non-invasive monitoring of DSL biomass and behaviour
- ✓ Solar-powered operation = zero carbon footprint for months-long missions

Scientific & Social Impact

- ✓ First system to quantify DSL carbon flux contribution to the biological pump
- ✓ Real-time illegal fishing detection — supports maritime law enforcement
- ✓ Provides climate scientists with continuous data on ocean CO₂ sequestration
- ✓ Informs fisheries management to protect ocean food security for 3B+ people

SDG Alignment

13 MESURES RELATIVES
À LA LUTTE CONTRE
LES CHANGEMENTS
CLIMATIQUES



Our carbon flux monitoring quantifies the ocean's CO₂ sequestration via the biological pump, directly supporting climate change mitigation and global carbon accounting.

12 CONSOMATION
ENT
PRODUCTION
RESPORABLES



Non-invasive sonar replaces destructive trawling. Eco-friendly acoustic design protects marine mammals. Illegal fishing detection preserves ocean biodiversity.

9 INDUSTRIE,
INNOVATION ET
INFRASTRUCTURE



Novel multi-frequency eco-sonar technology and ROV-buoy architecture represent a new paradigm in sustainable ocean sensing infrastructure.

17 PARTENARATS POUR
LA RÉALISATION
DES OBJECTIFS



System data shared with NOAA, CCAMLR, maritime authorities, and climate science community — enabling global cooperation on ocean monitoring and governance.

Scalability & Future Work

Phase 1 (Now)

Simulation validated. ROV & buoy CAD designs complete. Eco-sonar algorithm stack tested in MATLAB/Python.

Phase 2 (6 months)

Hardware prototype build. Tank & coastal trials. Dashboard MVP with live Sv feeds and alert notifications.

Phase 3 (1–2 years)

Open-ocean deployment at DSL sites (e.g., Mediterranean, Arabian Sea). Partnership with NOAA / oceanographic institutes.

Phase 4 (3–5 years)

Fleet of autonomous buoy-ROV units. Global DSL monitoring network. Carbon credit data pipeline for climate finance.

With More Time, We Would Add:

AI Species Classifier

Deep learning model on Sv frequency signatures for species-level classification at scale

AUV Autonomy

Replace tether with acoustic modem + onboard AI for fully autonomous adaptive survey paths

Acoustic Camera

Add 1.2 MHz imaging sonar to visually identify organisms and validate biomass estimates

Carbon Credit API

Standardised data format for verified ocean carbon sequestration metrics for carbon markets

Challenges & Limitations

Reduced Detection Range

⚠ *Eco-friendly source levels (175–180 dB) give 30–40% shorter detection range vs conventional sonar (200–210 dB).*

✓ Mitigation: Multi-frequency fusion and adaptive ping rate during migration windows recover most of the lost performance. Still viable for full DSL depth range.

Tether Length & Tangling

⚠ *ROV-buoy tether limits operational flexibility in strong currents and risks tangling in complex terrain.*

✓ Mitigation: Tether management reel on buoy. Future version will use acoustic modems for untethered AUV operation.

Surface Weather Dependency

⚠ *Solar buoy efficiency drops in prolonged overcast conditions or high sea states, limiting available power to the ROV.*

✓ Mitigation: Battery buffer sized for 5 days of autonomous operation. System enters low-power standby mode in adverse conditions.

Calibration Drift

⚠ *Long-duration deployments (weeks–months) cause transducer calibration drift, affecting biomass quantification accuracy.*

✓ Mitigation: ICES standard-sphere calibration scheduled every 14 days. Onboard drift correction algorithm applied in post-processing.

Not Yet Solved: Real Deployment

⚠ *Current results are simulation-validated. Physical prototype construction and open-ocean validation have not yet been completed.*

✓ Next Step: Phase 2 hardware build (6 months). Tank testing → coastal trials → open ocean deployment with scientific partner.

Not Yet Solved: Species ID Accuracy

⚠ *Frequency differencing identifies organism classes but cannot distinguish individual species within a class (e.g., krill vs copepods).*

✓ Next Step: Train deep learning classifier on multi-frequency Sv spectral signatures. Requires labelled ocean survey data.

Conclusion

Aqua Guardian presents a sustainable, non-invasive sonar system for monitoring the Deep Scattering Layer — delivering biomass estimation, carbon transport quantification, and illegal fishing detection at a fraction of the environmental cost of conventional systems, through a solar-powered buoy-ROV platform operating at echo-friendly source levels validated against NOAA marine mammal safety thresholds.